



Unit 3: Project Tracking and Monitoring

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Introduction to the Unit

In this unit, we will learn about the *Monitoring and Control Phase* of the project management life cycle. Upon completion of this unit, we will have a fair understanding of why control is an integral aspect of the project life cycle.

Some of the concepts that we will examine closely in this unit include the various activities performed in the *Monitoring and Control Phase*, the concept of *Performance Reporting* and *Issue Resolution* and the various types of *Measures* and *Metrics* used for project performance evaluation.

Objectives

Upon completion of this unit, you will be able to:

- Identify the key tasks in monitoring and control
- Identify the key activities in performance reporting
- Describe the issue management process
- Describe the importance of using measures and metrics
- Explain the Earned Value Management technique



Monitoring and Controlling a Project

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Objectives

Upon completion of this segment, you will be able to:

- Describe the process of monitoring and controlling a project
- Identify the key tasks in monitoring and control

Introduction

In the last segment, we discussed about the importance of review and its various types. In this segment, we will understand the role of monitoring and control in the life cycle of a project and the key tasks involved in monitoring and control.

Monitoring and control is the process, which oversees the entire gamut of tasks related to a project to ensure that the project remains within the pre-defined scope and the allocated budget and is delivered on time.

This process involves comparing actual performance of a project on the ground against the planned performance and taking appropriate corrective action to reach the desired outcome. This is a continuous process which is adopted across the life cycle of a project.

Key Tasks in Monitoring and Control

The key tasks involved in monitoring and control are:

- Scope verification and control,
- Schedule control,
- Cost control,
- Quality control,
- Controlling risks, and
- Reporting performance.

We will focus on the first four key tasks as 'controlling risks' and 'reporting performance' have been discussed later in other segments of this course.

The *Project Manager* has to adopt a basic principle of 'where we are, where we planned to be and how can we reach there?' to ensure that the project is monitored and controlled effectively. Figure 18 illustrates this principle.



Figure 18: Monitoring and Control – The Underlying Principle

Scope verification and control

The scope document created by the *Project Manager* and approved by the concerned *Stakeholders* stands as the reference for verifying scope. This scope document is directly related to work. Apart from actual work and deliverables, the scope document also refers to many other aspects of the project such as whether the project would include training, will it provide limited support, provision of documentation for the product and so on.

The *Project Manager* performs scope verification and control which identifies and manages all elements along with people, requirements and technology inside and outside the project that increase or decrease the project scope beyond the agreement.

The *Project Manager* has to keep an eye on all the resources, their utilisation, the technology being used as well as the agreed deliverables throughout the execution of the project. He/she has to ensure that resources used are within the agreed budget.

The *Project Manager* must also ensure that the technology being used for development is as envisioned earlier and agreed with the *Stakeholders*. Furthermore, he/she also needs to ensure that the deliverables being developed and delivered are as agreed and not more or less.

In case of any deviation, the *Project Manager* has to notify the *Stakeholders* and provide them information about the impact of such deviation. Any change to the project's scope has to come through proper channels, that is, through the 'Change Control Board' or the *Stakeholders* who can approve such changes based on a detailed analysis of its impact?

During the initiation of a project, the *Project Manager* has to record as to which *Stakeholders* hold authority to approve changes in the scope. Even he/she can go to the extent of distinguishing different *Stakeholders* based on priority and the severity of impact caused by such changes. For example, business-impacted decisions are approved by the *Sponsor* and technology-related changes are approved by the technical architect and so on.

Changes to the project scope can arise from any of the stakeholders such as the *Client*, *Sponsor*, *Project Manager* or a *Team Member*. Sometimes, changes are required mid-way in a project due to a gap in the initial scope documentation, requirement analysis or regulatory needs.



Project Management

The 'Change Control Board' plays a very important role in project scope management. The Change Control Board constitutes key *Stakeholders* who have the right to take decisions. These *Stakeholders* are either authorised by the *Sponsor* or provide details to the *Sponsor* to make decisions for major business impact changes.

The 'Change Control Board' is part of the change control process. The change control process simply refers to elevating the requested changes through appropriate channels and data to ensure that informed decisions are made. Figure 19 depicts the typical change control process. This process may vary slightly depending upon the organisation and the project.

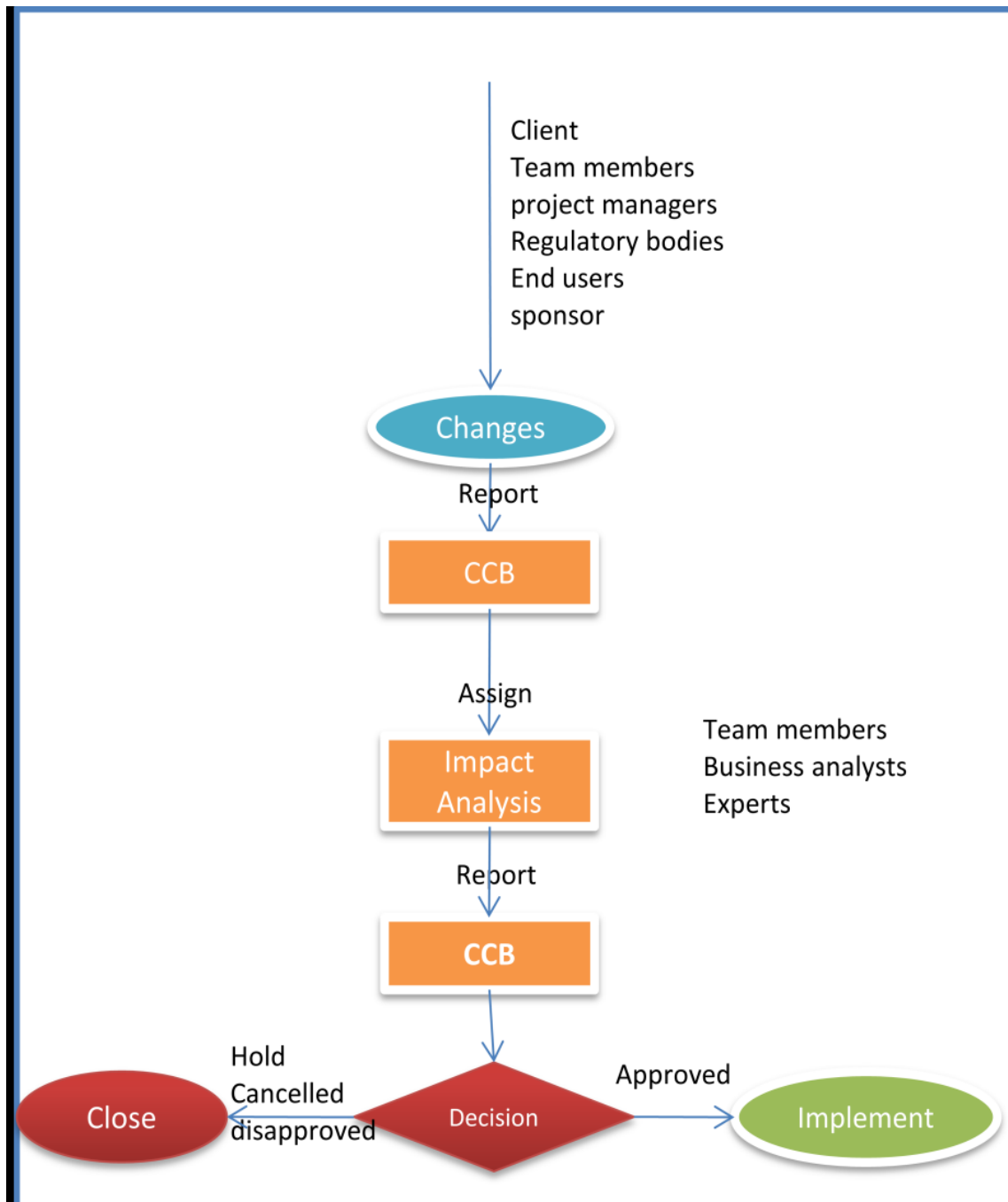


Figure 19: The Change Control Process



A very important part of the change control process is impact analysis and reporting.

The 'Change Control Board' has to know the impact of adopting a change in the project during this period. Does this impact cost, schedule, quality or any other?

This activity has to be done by trained people. Generally, business-related impacts such as legality, government policies and internal business decision impacts are analysed by the business analysis team from the client's side. Technical impact of such changes and the impact on schedule and cost are analysed by the *Project Team*. However, the *Project Team* has to provide all the details to support their claim on the impact.

The change control board decision-making team or the *Stakeholders* analyse these details and will give a 'go' or 'no-go' for such change requests.

Once the approval decision is received, the *Project Manager* has to update the plan and communicate all the details about the decision made. Moving further, the new plan will become the driving plan and the project progress needs to be tracked against it.

The change control process looks amazing on paper. However, when the *Project Team* members are working closely with the *Client* during the execution of the project, there will be occasions when the team members work on the *Client's* change requests without notifying anyone. These kind of scope changes which are incorporated in the project but not reported to the *Project Manager* or any *Stakeholders* lead to issues in the project at a later stage.

It is everybody's responsibility to adhere to the change control process in a project to ensure that the project is successful. The *Project Manager* has to make all efforts to make the team members and the *Stakeholders* aware of this process and elaborate on each individual's responsibility in the process.

Schedule control

Project schedule control is done purely by reviewing the project plan and tracking the progress. The *Project Manager* performs schedule control activities which involve influencing the factors that create schedule changes, to ensure that the changes made are beneficial to the project.

In many manufacturing industries, this schedule controlling activity is done by the project scheduler. The scheduler will have inputs about the resources such as machines and equipment and their performance. Based on the specific schedule goal that is to be achieved, the scheduler makes appropriate changes.

In the software industry, the *Project Manager* performs these actions explicitly. The *Project Manager* will know—from the inputs received from the resources—whether the project is behind schedule or ahead of the schedule. If the project is behind schedule then he looks for changing resources or identifying critical tasks which are holding up the progress. Allocation of more resources is done by the *Project Manager*.

Cost control

The budget for the project is identified and allocated during the initiation of the project by the *Project Manager* and the respective *Stakeholders*. A lot of research, analysis and estimates are done while



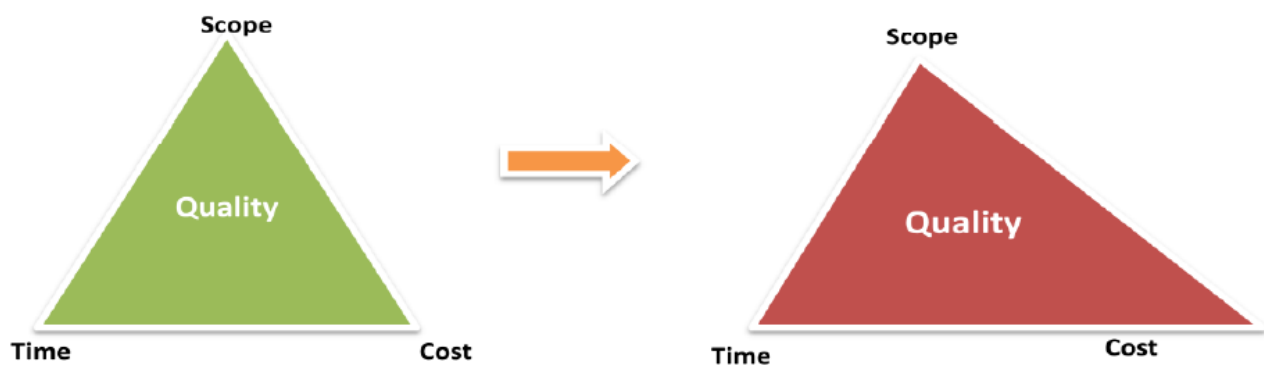
Project Management

allocating the budget. Adhering to the project budget at all times is essential to make profit from the project.

The *Project Manager* has to track all the actual costs against the planned or budgeted costs. The budget and tracking plan prepared by the *Project Manager* has to be time-based. It will help keep track of project costs at each phase and at each milestone.

It has to be verified whether the targeted costs at these phases are met or not. By utilising the work breakdown structure, the project budget has to be broken down properly and the associated cost estimated for each work package. In cases where associating cost at the work-package level is tedious, it can be done at a broader level.

The *Project Manager* has to make the team understand that the longer the project drags beyond the schedule, higher will be the costs. Even uncontrolled addition of requirements to the project can affect the cost. Therefore, project schedule and scope have to be controlled effectively to ensure that cost and budget are not adversely affected.



Similarly, scope change control and cost control also need to be effective. Scope control indirectly affects cost. If there is any increase in scope, the schedule will increase, which, in turn, could increase the project cost. The *Project Manager* has to communicate these points to all the *Stakeholders* and ensure that appropriate cost is recovered. The cost of the project can also increase due to poor quality of the project. This usually happens when defects or poor quality are detected at a later stage in the project life cycle.

The *Project Manager* has to regularly report to all *Stakeholders* on the utilisation of the budget and the forecast. Providing the forecast helps the respective *Stakeholders* to arrange for finance if the need arises.

Quality Control

Ideally, quality control is performed by a manager called the Quality Manager. The Quality Manager has to perform independently from the *Project Manager* or anybody who can influence his actions. In many projects, the *Project Manager* performs the role of a quality manager. This can affect the project quality.

Although the role of the *Project Manager* is to complete the project as per the agreed scope, schedule, cost and quality, in the urgency to meet timelines, the *Project Manager* might jeopardise the quality of the deliverables.



Project Management

The Quality Manager oversees all quality control activities which involve monitoring specific project results to determine if they comply with relevant quality standards set by the project or the organisation. In many organizations, quality is a separate function/department or group wherein they follow processes adopted by the organisation.

They would typically try to ensure that projects primarily follow the quality guidelines of the parent organisation and then the *Client* quality request, except in specific cases where the quality requirements of the *Client* are more stringent than parent organisation.

There is a quality team which operates outside the purview of the *Project Team* and adopts various techniques and tools to ensure the quality of the deliverables.

Quality management is a study in itself and works on a basic principle called the DMAIC technique, which was defined by Motorola as part of the Six Sigma management philosophy. This includes the following requirements:

Define quality: Define the quality expected by the project and/or its deliverables.

Measure quality: Measure the performance of quality.

Analyse quality: Analyse the deliverables and determine the root cause of poor quality, and determine how it can be improved to achieve the desired quality levels.

Improve quality: Improve the quality of deliverables by attacking the root cause and identify solutions that are easier to implement.

Control quality: Observe the implemented improvements for success, update records regularly and maintain this routine.



Figure 20: The DMAIC Technique



Project Management

Quality control is inspection driven and is all about identifying and rectifying mistakes made by the project before it reaches the end customer. The *Project Manager* and the team have to work to ensure that all the work is accurate, within scope and meets the objectives set forth by the *Client*. Quality control is time consuming. When defects are found, it takes a long time to fix them in most cases.

The *Project Manager* has to schedule a proper timeframe for the quality control team to perform their role as well as for the *Project Team* to fix the errors. Although this check and fix is not easy to estimate, the *Project Manager* has to ensure that his team is aware of this and that they work diligently while producing the deliverables so that any adverse impact on the project can be minimised as much as possible.

When the quality control team identifies defects in a product, the *Project Manager* can leverage many of the available tools to identify the root cause of these issues and resolve them.

Some of quality control tools used for performing root-cause analysis of quality defects are as follows:

- Ishikawa diagram
- Pareto Chart
- Control charts
- Cost-benefit analysis

If the defects found are straightforward and are easy to fix, it is not much of a problem. However, if the number of defect is more, a proper analysis becomes necessary to root out the problem. Instead of fixing each issue, it is cost effective and saves time to identify the root cause of the problem and address it.

In this way, monitoring and control proves to be an important aspect of the project life cycle and its key tasks can go a long way in ensuring that the project does not meander away from its planned trajectory and, if it does, it is brought back to the course by following the recommended processes.

Summary

In this segment, you learned that:

- monitoring and control is the process, which oversees the entire gamut of tasks related to a project to ensure that the project remains within the pre-defined scope and the allocated budget and is delivered on time; and
- the key tasks in monitoring and control are scope verification and control, schedule control, cost control, quality control, controlling risks and reporting performance.



Performance Reporting

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1. Objectives
2. Introduction
3. Key Activities in Performance Reporting
4. Types of Performance Reports
5. Summary

Objectives

Upon completion of this segment, you will be able to:

- Explain the process of performance reporting
- Identify the key activities in performance reporting
- Identify the types of performance reports
- Describe how reports are generated

Introduction

In the previous segment, we learned about the basic principles of the independent process of monitoring and control and the several key task of monitoring and control. In this segment—going further on the topic of monitoring and control—we will focus on the concept of performance reporting.

A project cannot be executed without understanding its status and progress. Project work requires on-going measurements and evaluations to ensure everything is on track and within time, cost, quality and scope.

Stakeholders of a project need to understand where the project is with respect to their commitments? The *Project Manager*—while tracking the progress of project—has to share the same with all the *Stakeholders*.

Performance reporting is the process of collecting and distributing performance information such as status reports, forecasts, progress measurements and variances to the respective *Stakeholders*.

The audience for receiving performance reports, or for that matter, any report is based on the varied needs of the *Stakeholders*.

The *Project Manager* has to understand the needs of different stakeholders, group them accordingly and share reports.

The *Sponsor* of a project may only need an overview of the status of the project which can say how much percentage of work is completed, what is the incurred cost, remaining budget and how is the project doing with respect to the committed delivery dates. His needs can be met by a simple consolidated report.



Other *Stakeholders*, who are interested in understanding the day-to-day or the weekly progress of the project, may need more detailed reports. Such reports may need to have a list of deliverables, their due dates, percentage completion of each deliverable's work and so on. There could be another set of financial *Stakeholders* of a project who needs to have a closer look at the expenses of the project, how the budget is being utilised and how much of budgeted amount needs to be approved before they start work on the next milestone and so on.

Different *Stakeholders* have different requirements and the *Project Manager* has to follow the process of performance reporting to ensure that all such data is captured properly and arranged so it can be distributed in the shape of reports.

Key Activities in Performance Reporting

The performance reporting process involves some key activities, which include

- engaging with the *Stakeholders*,
- identifying and prioritising performance indicators,
- collecting and compiling, and
- preparing and publishing.

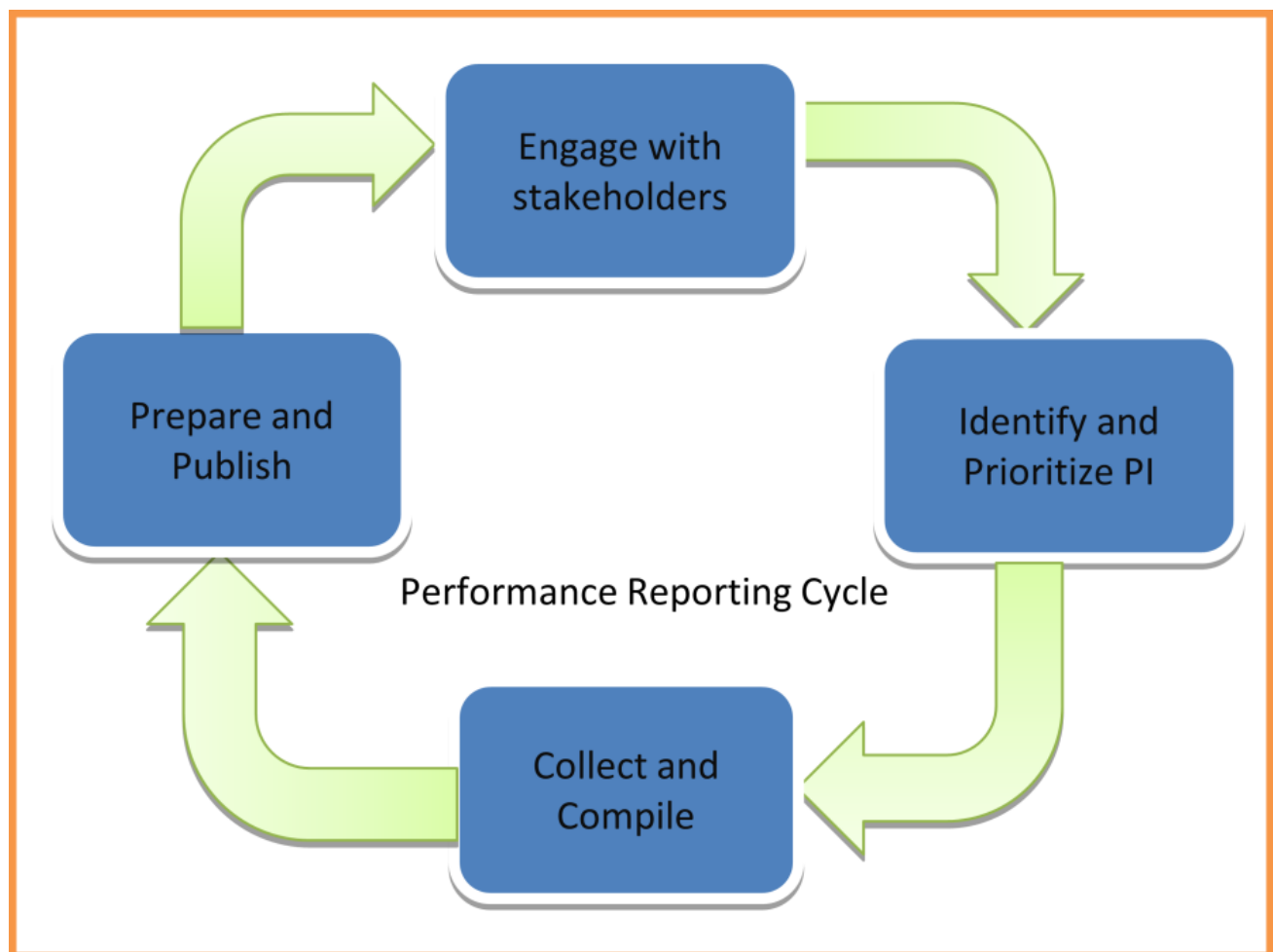


Figure 21: Activities in Performance Reporting



Engaging with the *Stakeholders*

After identifying the *Stakeholders*, it is essential to hold discussions with them and identify their expectations. Their expectations have to be clearly understood and documented. This can be done through discussions, meetings or by sending emails and obtaining their response to a set of questions. This goal is achieved at the initial stage of project where the expectations of all the *Stakeholders* are collected and documented.

Identifying and prioritising performance indicators

Performance indicators are measures which have to be collected in the project and reported. For example, in a manufacturing unit, the performance indicator of one *Stakeholder* could be to look at the 'number of rejected units'. Such indicator expectations have to be gathered from the *Stakeholders* and then recorded and analysed. The purpose of the analysis of measures is to prioritise them and plan for their collection.

Collecting and Compiling

This is a long-term activity, which involves regular collection and compilation of data. Data can be collected in various ways and from various sources. Certain data may be available from the project plan, which is under tracking or from the reports prepared by the quality team, group/team leads or directly from the team members or other resources. Data can even be collected during meetings with the team.

Once the report is collected, it has to be compiled properly and, if required, proper matrices have to be derived. A metric is a calculated or composite indicator based on one or two measures. We will examine the metrics in more detail in a later segment.

Preparing and publishing

After the measures and metrics data have been collected, they need to be documented in a format, which makes sense and must be reported to the respective *Stakeholders*. Many standard tools such as Excel, PowerPoint can be used for this purpose. Even custom-made dashboards which provide these details in specific formats in one screen can be used. This depends on the *Project Manager* and the *Stakeholder's* expectations.

Types of Performance Reports

In general, based on the type of report being shared, performance reports can be classified as:

- Progress reports
- Forecast reports
- Status reports
- Variance reports



Progress reports

Progress reports are used to describe the trend from past to the present. These reports are generally graphical and illustrate trends. Figure 22 displays a sample cost progress from March to July.

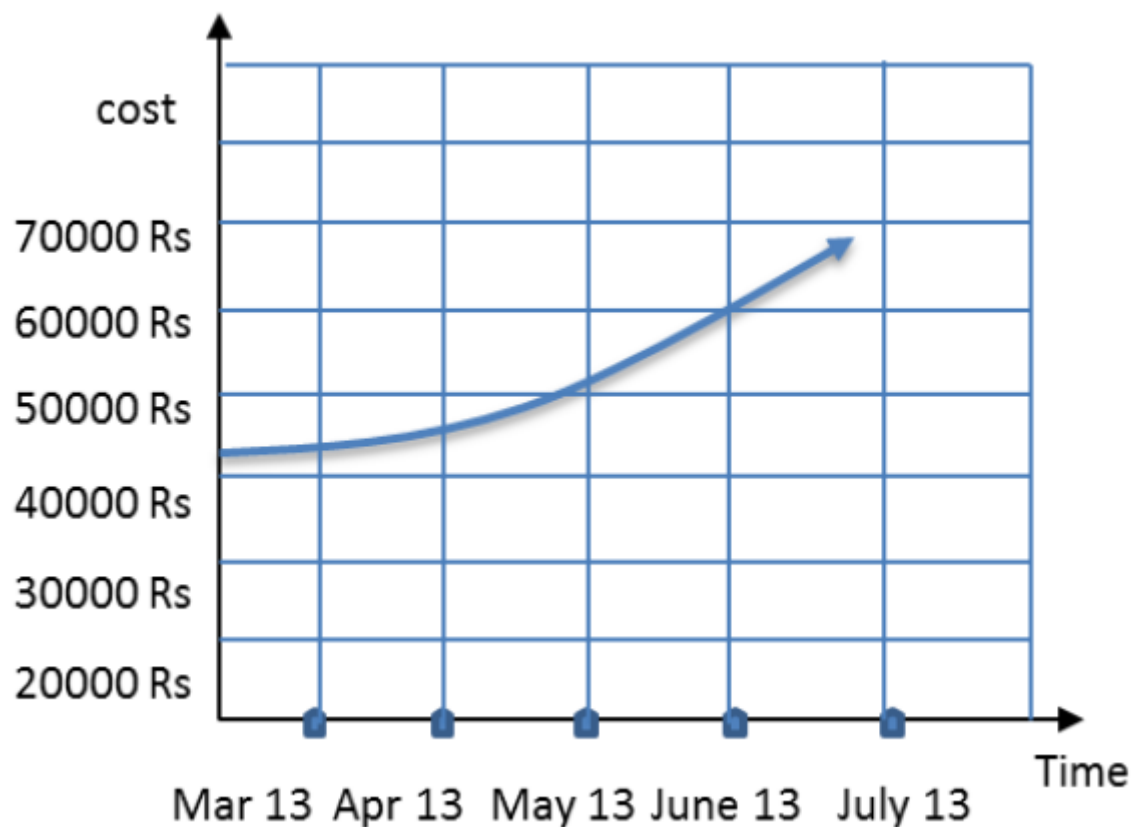


Figure 22: Cost Progress

Forecast reports

Forecast reports describe future trends. To calculate a future trend, measures collected so far will not be sufficient. Future trends can be depicted either through experience or by using scientifically proven formulae. If a machine is consistently producing a product for the last few days then—under ideal conditions and given the assumption that the machine will work the same way for the next few weeks—the future trend of production can be arrived at.

If the influence of available raw materials and planned downtime is also considered, it becomes difficult to straight away predict the future trend. In such cases, certain formulae need to be used which must be approved by the organisation. Forecasts provided by such reports help to steer the project properly.

In figure 23, the forecast is shown by a green-dotted progress line. The forecast appears to be constant from July onwards.



All forecast reports need not be in a graphical format. They can also be represented in a tabular format.

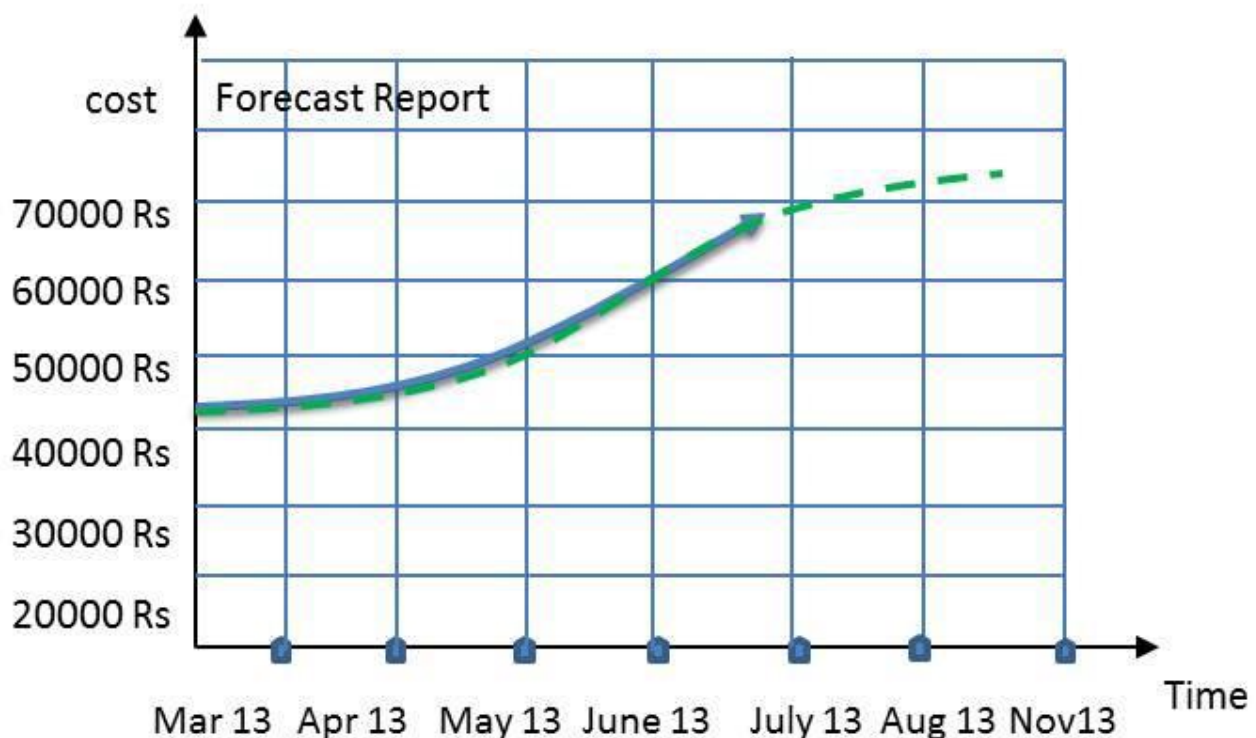


Figure 23: Graphical Representation of a Forecast Report

Status Reports

Status reports describe the actual status on the ground. They provide details about the status of the deliverables—or items of interest—at the time of reporting. Generally, the trend across organisations is to provide a weekly or bi-weekly status report.

Again, based on the type of project and reporting requirements of *Stakeholders*, this communication can happen as required. In the case of some projects, daily reports are submitted at the fag end of the project period.

Status reports can, again, be either graphical or textual. Most often, the status reports provide data pertaining to the current week and performance in the past period. The report can even be directly fetched from the project plan under tracking.

Colour indicators are also used in status reports to mention if the current status is good, bad or worse. Colour coding helps *Stakeholders* and project managers to concentrate the discussion on high-priority items and take prompt action.

The main difference between progress and status reports is that a progress report gives quick information on where the overall project is; for instance, whether 50% complete, 60% complete and so on. On the other hand, a status report is very detailed and is more specific to a particular period and not the whole project.



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Figure 24 displays a typical status report.

Project Status Report		Week 10							
Project Name		ManipalAccounts							
Project Manager		Vishwesh							
Current Date		21-Mar-13							
Sl No	Deliverables	Effort (hrs)	Planned Start Date	Planned Finish Date	Actual Start Date	Actual Finish Date	%ge Completion	Current Stat	
1	Peer Review Design 4	8	20-Mar-13	21-Mar-13	20-Mar-13	21-Mar-13	100%		
2	Peer Review Design 5	8	20-Mar-13	21-Mar-13	20-Mar-13	21-Mar-13	100%		
3	Build Deliverable 1	30	15-Mar-13	18-Mar-13	15-Mar-13	18-Mar-13	100%		
4	Build Deliverable 2	45	15-Mar-13	21-Mar-13	15-Mar-13	NA	90%		
5	Build Deliverable 3	60	15-Mar-13	25-Mar-13	15-Mar-13	NA	60%		
6	Build Deliverable 3	45	21-Mar-13	25-Mar-13	21-Mar-13	NA	0%		
7	Build Deliverable 4	45	21-Mar-13	25-Mar-13	21-Mar-13	NA	0%		

Figure 24: Status Report

Variance Reports

Variance reports provide details about the differences between the initially planned data and the current data. These are also called variance-analysis reports. These help to gauge the performance by comparing one set of data with another.

In project management, there are different views on variance reports, mainly, estimate to planned, planned to actual and estimate to actual.

Estimate to planned: This is the difference between what is actually estimated initially and sold to the *Client* against the planned values. It could be effort of work in hours, cost or schedule. Ideally, the estimate and planned has to be same.

However, it is possible that the estimate is different from the planned when they are all dealt by different people with different needs. This variance is seen by top management or senior *Stakeholders* of the organisation. It helps them to adjust and quote their prices appropriately to the *Client* the next time.

Planned to actual: This is the most common variance which the *Project Manager* reports to all *Stakeholders*. It is the difference between planned scope, cost and schedule against the actual values on ground, till date. The *Project Manager* can often use this report on his/her own to ensure that the project is executed properly.

If there are large differences between planned and actual, the *Project Manager* needs to find out why and work with team to ensure that the gap is minimised. He can set up guidelines or use the organisational guidelines on when to escalate variances.

For instance if the variance in cost is between 0 - +/-5%, then it need not be escalated. However, if the variance is above 5% then it needs to be escalated to the appropriate *Stakeholders*.

In case the variances continue for a longer period, the *Project Manager* needs to adopt different techniques to identify the root cause of such variance and resolve it.



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Some of the commonly employed techniques include brainstorming, the Ishikawa diagram, the 80-20 rule and so on.

Estimate to actual: This is a comparison of what was estimated earlier and quoted for the work/project against the actual costs. This is also an important comparison. If the difference between estimated and actual is positively high then it leads to profit but if it is less then it results in losses. However, if the difference is high, Management Leads need to take caution to ensure that the estimate and actual are closer.

Again, by analysing the difference properly—what changed, why and when—corrective action can be taken while there is still time.

Most of these reports are presented by the *Project Manager* to the *Stakeholders*. However, looking at the above reports individually can lead to many questions. For instance, while analysing a status report, it may be found that there is cost difference of USD 100. In such a situation, it becomes difficult to come to conclusions with just one value.

However, if it is combined with the trend for the past couple of weeks, or for the whole duration, it can lead to some informed decisions.

Does that mean that the *Project Manager* has to combine the current status report, progress report, and variance and forecast reports?

As mentioned earlier, it depends on the *Stakeholder's* expectations. It is possible to overload all the *Stakeholders* with information but it may take a long time to explain things. In many cases, it may not mean much to the *Stakeholder*. To save the *Project Manager* and the *Stakeholders'* time, reports need to be classified or clubbed as required and submitted.

Furthermore, while building and presenting reports, the *Project Manager* has to see how the reports should be presented. He/she could do it in a face-to-face meeting with the *Stakeholders*, over a video conference or over phone/email. The report can also be altered as per the medium of communication.

Many organisations gather key measures from the project and generate reports using tools which support project management and reporting. For a small project, it would be easier for the *Project Manager* to generate all such reports. If the project is complex, critical and/or huge then the *Project Manager* will have less time to generate these reports.

However, providing status reports to stakeholders is a key role of the *Project Manager* and it has to be performed. That is where, standard software programs come in handy because they help build these reports, verify data and use for sharing.

If the project management software is used across the organisation and even the client *Stakeholders* are given permission, they can view the reports themselves. The *Project Manager*, however, has to ensure that accurate data is periodically updated to the project management software.

Last but not least, it is important to note that based on the mode of receiving reports, they can be again classified as 'Push' or 'Pull' reports.

In 'Push' type, the reports are pushed by the *Project Manager* to all *Stakeholders*. If the *Stakeholders'* numbers are large, then a status report has to be pulled by the *Stakeholders*. For



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instance news channels, newspapers and announcements wherein the *Stakeholders* are responsible for pulling the information they want.

In this way, performance reporting serves an important function in the project life cycle and helps the various *Stakeholders* stay abreast of all aspects of project performance.

Summary

In this segment, you learned that:

- performance reporting is the process of collecting and distributing performance information
- the key activities of performance reporting include engaging with the *Stakeholders*, identifying and prioritising performance indicators, collecting and compiling and preparing and publishing.
- performance reports can be classified as, progress reports, forecast reports, status reports and variance reports.



Resolving Issues

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3. The Issue Management Process
4. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe the issue resolution process
- Identify the steps in the issue resolution process

Introduction

In the last segment, we learned about the concept of progress reporting and the various types of reports used in this function. In this segment, we will focus on resolving issues that arise during the life cycle of a project.

In any project—almost always—problems or questions arise due to which either the complete project progress may stop or the work related to them gets affected. These problems or challenges are nothing but project-related issues which can be solved by adopting the issue resolution process.

Before we discuss the issue resolution process, let us first understand the difference between risks and issues. In case of risks, there is a certain feeling that things can go wrong and you plan to respond if that happens. An issue is less predictable and can arise with no prior warning or notice.

For example, the unavailability of the right tool in the market to perform a certain activity can be regarded as an identified risk. However, if that tool is available and if it breaks down while we are working with it, and we are unable to move ahead until a replacement arrives—which could take up to a week—would be an issue.

With respect to time, a risk is there in the future and an issue is there in the present.

Risks identified earlier—when not managed properly—can turn into issues. If an elaborate list of risks is made with triggers attached to each one then they must be properly monitored. When the triggers of risks are neglected, then eventually, the risks turn into issues. Proper risk management and control is important to reduce issues in a project.

Most of the issues are unexpected. How do you deal with them properly?

Project Managers have to be prepared for effectively handling issues at all times. The *Project Manager* must adopt a proper issue resolution process when the project is initiated. All the *Stakeholders* of the project have to be made aware of this process and its existence and availability.

The right owners and points of contact for such processes have to be published regularly. Generally, people would not look into such processes and try to understand them unless they are



pushed into a critical position by someone or by their work. The *Project Manager* can regularly send reminders or tips—on such processes—to remind the team about these processes and the way they work.

The Issue Management Process

The issue management process is all about what to do with issues and to ensure that they are resolved properly.

By using the issue management process, the *Project Manager* will be able to:

- assign actions to team members to resolve issues,
- assign roles and responsibilities to manage issues,
- monitor the outcome of actions taken,
- report the status of issues, and
- construct a database of knowledge to resolve similar issues in the future.

The process of issue management involves:

- identification of issues,
- evaluation and documentation,
- determining actions for resolution,
- monitoring and controlling the actions, and
- closure of issues.

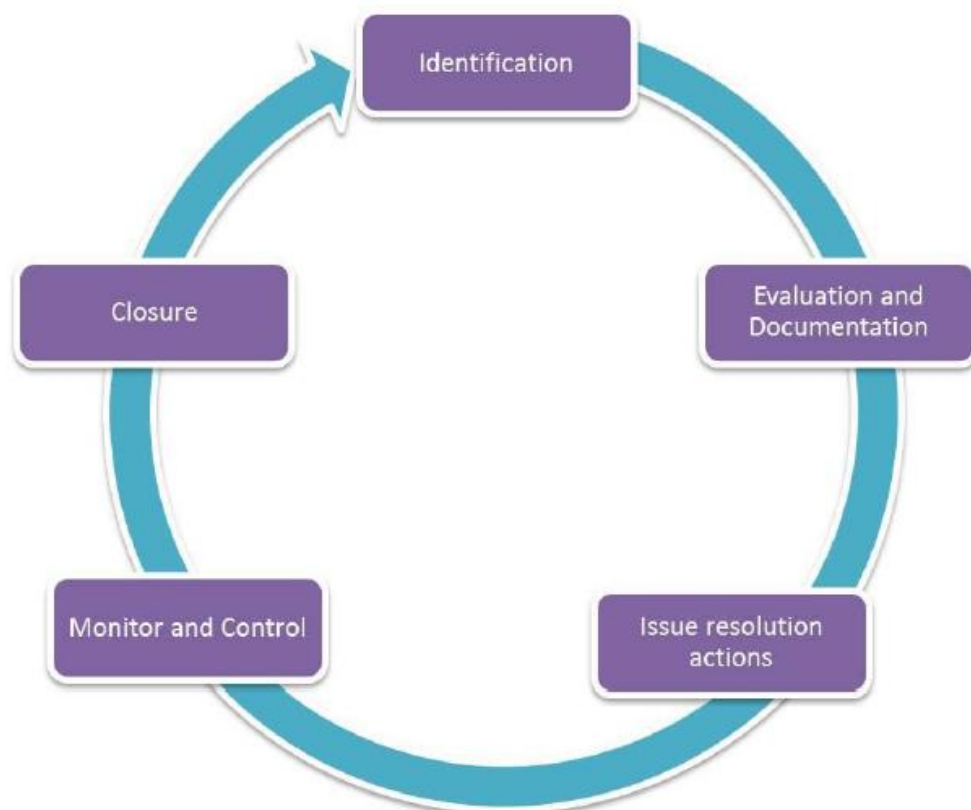


Figure 25: The Issue Management Process



Identification of issues

The client, customer or any *Project Team Member* can raise a project related issue at any time. As soon as the issue arises, the concerned person has to inform the *Project Manager* or the relevant Lead or point of contact about the issue.

Evaluation and documentation

The *Project Manager* should review the issue himself/herself or with some other member who has knowledge about the issue. If the *Project Manager* does not have proper data about the issue, then he/she needs to obtain it from a direct report. He must get all the relevant information such as when, how and in what circumstance did the issue arise.

Then it has to be analysed whether it is an issue or not and how it impacts the project. The decision should be based on its likely impact on deliverables, schedule, cost, scope, motivation, quality and any other aspect of the project.

All issues mentioned as issues by any *Stakeholder* have to be recorded in a formal document known as the issue log.

During a detailed analysis, the issue has to be categorised based on the categorisation criteria that has been set forth. The priority of the issue has to be set as very high, high, medium or low. There can also be specific owners assigned for each category of issues.

Determining actions for resolution

After investigating the issue with the appropriate team, the *Project Manager* can decide to take actions, which may include:

- closing the issue in the issue log if there is no impact on the project;
- raising a change request and sending it to the Change Control Board for a thorough analysis followed by a decision;
- raising the issue as a project risk, if it is likely to impact the project in the future;
- assigning the issue to its owner for a resolution; and
- documenting the action required to resolve the issue and providing its timeline.

Monitoring and controlling the issue

Once the issue is assigned for action, the *Project Manager* has to monitor it closely to ensure that required action is taken on it. Regular status updates need to be taken and recorded and also communicated to all stakeholders based on the impact and severity of the issue.

It may not be necessary that the details of all the issues be communicated to all the *Stakeholders*. The status can be shared only with the *Stakeholders* whose interests are directly impacted and who have been authorised to know about it.

After obtaining confirmation from the issue owner that the issue is resolved, proper verification has to be done to ensure that the required action is properly implemented.



Closure of the issue

When the issue is verified and closed, the *Project Manager* has to ensure that the issue is formally closed in the issue log with all the relevant details.

Issue log

Issues which are known as problems, conflicts, inconsistencies or show stoppers have to be recorded. The issue log is a tool for recording and communicating what is happening in the project. This record ensures that issues were actually raised, analysed and resolved during the course of the project.

An issue log has to be created by the *Project Manager* in the initial stage of the project when the issue management plan is drawn.

This log acts as a:

- reliable method for the team to raise issues,
- tracking tool for Leads/Project Managers,
- tool for analysing and recording details of all issues at one place, and
- knowledge base for the team to quickly act on similar kinds of issues based on the issue resolution details documented in it.

The issue log can be a spreadsheet sheet or any such similar software. Figure shows a typical issue log.

Issue Log												
Issue Reference No	Issue Type	Identified by	Occurred On	Short Description	Description	Impact	Priority [P1,P2,P3]	Planned Response Action	Owner	Response Action	Issue Event log	Status

Figure 26: Issue Log

An issue log can contain the following information:

Issue reference no.: This is a unique number for each issue. For example, Prj_Issue_01.

Issue type: It defines the categories of issues as defined by the project. For example, technical or functional or related to human resource, equipment, power, vendors and so on.

Identified by: This refers to the person name or ID who identified and reported the issue.

Occurred on: This indicates when the issue occurred; generally, the date and time.

Short description: This is a brief description of the issue and is helpful for glancing over the issues if the list is too long.

Description: This is a detailed description of the issue, which explains when, how and what happened.

Impact: Post analysis, the impact of the issue is determined and documented here. It can be categorised as high, medium or low or as 1, 2 and 3.



Priority: Based on the investigation and analysis, priority of the issue is set by the *Project Manager*. It could be high priority, medium priority or low priority or P1, P2 and P3.

Planned response action: Based on the decision, the planned response action is documented here.

Owner: The *Project Manager* assigns an owner for each issue, who owns it—to ensure that the planned response action is implemented—and updates the response to the action, on time.

Response action: This describes the action performed on the issue to resolve it.

Issue event log: This is documented by the owner or the *Project Manager* on the step-wise action taken to resolve the issue.

Example:

Mar 10: Assigned to xyz and all instructions provided by the owner.
Mar 11: XYZ is working on the machine and is following instructions.
Mar 12: Work completed as per instructions and testing begun.
Mar 13: Testing in progress.

Status: This provides the status of the issue in discussion. It is updated by the respective owner of this log.

The status can be: Resolved, Implementing, Investigating, Escalated or Open.

In this way, the issue resolution process proves itself to be a valuable tool for the *Project Manager* to identify and resolve the issues that invariably crop up during the life cycle of a project.

Summary

In this segment, you learned that:

- issues are different from risks in that the former is less predictable and occurs in the present whereas the latter is more predictable and occurs in the future;
- the issue management process deals with what to do with issues and to ensure that they are resolved properly; and
- there are five stages in the issue resolution process, namely identification of issues, evaluation and documentation, determining actions for resolution, monitoring and controlling the actions and closure of issues.



Measures and Metrics

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Objectives

Upon completion of this segment, you will be able to:

- Describe measures and metrics as methods of performance evaluation
- Explain the metrics process life cycle
- Explain the concept of Earned Value Management (EVM)
- Identify the reasons for using metrics

Introduction

In the last segment, we learned about resolving issues that arise during the life cycle of a project and the various steps involved in the issue resolution process. In this segment, we will focus on the importance of measures and metrics and how they are used to measure the project performance.

Measures and Metrics

Measure is a value that is quantified against a standard at that point in time. It is actually a mere number associated with certain attributes such as currency value, number of units and so on.

A single measure value has little value without a relevant context. But it is important to arrive at a proper meaning.

For instance, the total number of deliverables produced in the second week of January is 20. How do we know if this number is good or bad for a given project?

If the number of deliverables produced is tracked on a weekly basis and it is found that this week, the output is higher than previous, then it is a good sign.

A metric is a degree to which a particular subject possesses the quality that is being measured.

Metrics are derived based on two or more measures.

Another way of describing measures and metrics is:

A measure is an indication of the size, quantity, amount or dimension of a particular attribute of a product or process.



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A metric is a measurement of the degree to which any attribute belongs to a system, product or a process.

For example, the number of errors per person hours would be a metric. It is based on a formula where parameters are measures.

There is no single set of metrics that can be used by all. The appropriate set of measures depends on the organisation's strategy, technology, maturity, industry and environment.

Project matrices focus on various factors such as schedule, scope, contracts, cost/budget, status, stakeholder involvement, resources, quality, risk and change management.

Some of the project-related metrics are:

- **Training hours spent:** number of training hours spent per project team member.
- **Overdue tasks:** number of tasks that were started but not finished on time.
- **Scope change requests:** number of changes requested during the course of a project.
- **Earned value:** Budgeted cost of work performed.

The Metrics Process Life Cycle



Figure 27: Metrics – Life Cycle



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The metrics life cycle follows a very simple process of define, measure, analyse, action and improve/remove.

Define

Define metrics that are required for the project/organisation. Identify only those metrics that will help. It may be an overkill if the metrics chosen are non-beneficial.

Measure

Start measuring the data. Capture data.

Analyse

Analyse the captured measures and metrics data.

Action

Post the analysis, based on the outcome, take action to ensure that corrections are made to the project.

Improve or Remove

This is the monitoring part of the cycle wherein it is essential to continue to cross-check to ensure that the metrics used are correct and beneficial. If the metrics need any tweaking or adjustment, it must be done. Otherwise they should be removed.

Earned Value Management

Now, let us look at one of the most common measures and metrics captured in a project under a technique called Earned Value Management (EVM).

Earned value management or EVM is a method used by Project Managers to measure the project performance. EVM can be used on cost and schedule control. It can also be used for forecasting the same. In this segment, we will look at how cost and schedule variances are identified and controlled.

EVM measures

EVM consists of the following primary data points that need to be captured. Each data point value is based on the time/date an EVM is measured.

Budget at Completion (BAC): Total cost of the project.

Budgeted Cost of Work Scheduled (BCWS) or Planned Value (PV): This is the amount expressed in currency value or hours of work to be performed as per the scheduled plan.

$PV = BAC \times \% \text{ of planned work.}$

Budgeted Cost for Work Performed (BCWP)/Earned Value (EV): The amount expressed in currency value or hours on the actual worked performed.



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$EV = BAC \times \% \text{ of actual work.}$

Actual Cost of Work Performed (ACWP)/Actual Cost (AC): The sum of all costs or hours actually accrued for a task to date or till the time of measure.

Example: We are supposed to complete USD 400 of work today and have completed USD 600 of work already. Actually incurred cost or paid till now for work done is USD 500.

BCWP or EV = USD 600

BCWS or PV = USD 400

ACWP or AC = USD 500

Based on the above measures that have been captured, we can derive metrics which provide more details.

Cost Variances (CV)

Cost Variance (CV): Are we under budget or over budget?

$CV = EV - AC$

Negative value of CV represents that we are over budget. It means that we are spending more and earning less. Hence, there may be need for more budget allocation.

Positive value of CV represents that we are under budget. It means we are spending less and earning more. We are under utilising the budget but are still earning more.

Schedule Variances (SV)

Schedule Variance (SV): How much ahead or behind the schedule are we?

$SV = EV - PV$

A negative value represents that the project is behind schedule.

A positive value represents that the project is ahead of schedule

Let us take a typical example and find these variances.

You are a *Project Manager* managing the construction of a 2 km-long road. The budget allocated for this work USD 80,000 and is expected to be completed in 10 days. You have already started work and are on the fifth day of the project.

You are taking an account of the progress and find that you have already spent USD 55000 and have completed 50% of construction which is 1KM.

Are you behind or ahead of schedule and under or over budgeted at this moment?

BAC = 80000

$PV = BAC \times \% \text{ of planned work} = 80000 \times 50\% = 40000$

$EV = BAC \times \% \text{ of actual work} = 80000 \times 50\% = 40000$

AC = Actual cost = USD 55000



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$$SV = EV - PV = 40000 - 40000 = 0$$

$$CV = EV - AC = 40000 - 55000 = -15000$$

Schedule: The project is on schedule with respect to the work being completed.

Cost: The project is under budget by USD 15000. Spent USD 15000 extra to achieve the above schedule or may require USD 15000 extra by the end of the project if it continues the same way without further changes.

As mentioned above, above measures along with metrics can also be used to forecast the performance. We will keep it aside as part of a detailed study on EVM.

Why Use Metrics?

While it can take time to create and report metrics, there are numerous reasons why an organisation should take advantage of them. One of the best-known reasons for measuring project metrics is to determine the status or the 'health' of a particular project.

Metrics can also give important objective information about certain details of a project, and along similar lines, the metrics can be a source of imperative information for data control. Specifically, metrics can also help with:

- risk mitigation,
- issue resolution,
- project control,
- team performance management,
- client expectations management,
- adjustment of development plans,
- prevent scope creeping while executing, and
- better accountability.

Metrics can also help in understanding customer satisfaction, team satisfaction measurement, resource management efficiency, identifying and understanding bottlenecks in the systems development life cycle and managing the software quality. Additionally, IT metrics can control and measure IT processes such as production operations, software development and security and application maintenance.

Some of the other specific management areas that can benefit from metrics include gaining the ability to quantify staff activities and outsourcing of work.

Why metrics are useful

Metrics are extremely useful as these measurements can be used in every stage of the management process.

During the planning phase of a project, metrics can be used to plan the allocation of resources, budgets, schedules and cost estimations. Further, metrics can be used to discover information about the status and quality of projects.

They can also identify specific areas that require improvement. However, in order to be useful, metrics must have the following three characteristics:



Project Management

- **Functionality:** The metrics must measure exactly what numbers they are supposed to measure.
- **Cost-effectiveness:** The metrics must deliver more value than the cost of measuring and analysing the metrics information.
- **Consistency:** The metrics must be reliable throughout the company operations of the organisation

Of course, the metrics that an organisation chooses to use would depend on the particular project that it is working on and depends on what exactly their target is. Also, before choosing the project metrics, the project objectives must be identified clearly.

Once the objective is certain, the appropriate metrics can be chosen. The metrics that are required to measure the progress of a project are different from the metrics that are required to solve a crisis.

At this time, it is also important to figure out exactly which are the questions that the organisation wants answered

An organisation should not choose a particular metric based on the fact that a particular metric is used often.

At this point, one may ask, 'Why?'

Simply, because one size does not fit all and also because the fact that a particular metric is utilised does not mean that this metric will be a helpful measure for the specific project or the organisation.

Thus, by asking the right questions at this stage, the organisation will be able to choose the metrics that will help it find the answers that it seeks. After this, it must be able to collect, measure, and interpret this data in an effective manner.

In this way, if the project metrics are developed and implemented in the proper manner, they can go a long way in helping the project as well as the organisation.

Summary

In this segment, you learned that:

- a measure is an indication of the size, quantity, amount or dimension of a particular attribute of a product or process;
- a metric is a measurement of the degree to which any attribute belongs to a system, product or a process;
- the metrics life cycle follows a very simple process of defining, measuring, analysing, taking action and improving/removing;
- one of the most common measures and metrics captured in a project is EVM; and
- there are many reasons for measuring project metrics, most important of which is to determine the status or the 'health' of a particular project.